Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India
(Autonomous College Affiliated to University of Mumbai)

Mid Semester Examination
September -2024

Max. Marks: 30 | **Duration: 01 hour**
Class: B.Tech. | **Semester: V**
Course Code: CS-304 | **Branch: Computer Engineering**
Name of the Course: Distributed Computing

Instructions:
(1) All Questions are Compulsory.
(2) Draw neat diagrams.
(3) Assume suitable data if necessary.

Q. No.	QUESTIONS	Max. Marks
Q.1	What are the major design issues of distributed systems? Which are the different dimensions to measure the scalability of a distributed system? What are the scalability problems and solutions/techniques? Explain with example.	06
Q.2	A multicomputer with 256 CPUs is organized as 16 × 16 grid. What is the worst case delay (in hops) that a message might have to take?	04
Q.3	With persistent communication, a receiver generally has its own local buffer where messages can be stored when the receiver is not executing. To create such a buffer, we may need to specify its size. Give an argument why this is preferable, as well as one against specification of the size.	05
Q.4	An experimental file server is up 3/4 of the time and down 1/4 of the time due to bugs. How many times does this file server have to be replicated to give an availability of at least 99%? Justify the answer.	05
Q.5	Why transient synchronous communication has inherent scalability problems, and how these could be solved.	05
Q.6	What are the main reliability issues in designing a message-passing system? Describe a suitable mechanism for handling each of these issues.	05